

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 6D nr. 30.11

$$\begin{aligned} & \sum_{n=2}^{\infty} \left(\frac{|\sin 2^n|}{n^3} + \frac{5 \ln n}{n^3} \right) \\ &= \sum_{n=2}^{\infty} \frac{|\sin 2^n|}{n^3} + 5 \sum_{n=2}^{\infty} \frac{\ln n}{n^3} \\ & \sum_{n=2}^{\infty} \frac{1}{n^3} \text{ majorant convergent over } \sum_{n=2}^{\infty} \frac{|\sin 2^n|}{n^3} \\ & \sum_{n=2}^{\infty} \frac{\ln n}{n^3} \Rightarrow \text{convergent, al in een eerdere oefeningen bewezen} \\ & \Rightarrow 5 \times \text{convergent} = \text{convergent} \\ & \Rightarrow \text{convergent} + \text{convergent} = \text{convergent} \end{aligned}$$

References